

Universidad Nacional Autónoma de México
Faculty of Engineering
Data Mining

Practice 3.6
Convolutional Neural Networks (CNN)
Facial Emotion Recognition with FER-2013

Team 10
Prof. M.C. Jose Roberto Olvera Lopez

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1 Purpose of the Practice

1.1 Background: The Original Study

In 2020, Sturm et al. (UCSF) published a randomized controlled trial with 60 older adults (ages 60–90) investigating whether brief walks designed to cultivate **awe** could improve emotional well-being [1]. Over eight weeks, participants took three *selfies* per walk — 1,184 photos in total — and completed emotional state surveys at the end of each session.

To analyze behavioral changes in the photographs, the authors used **MATLAB Image Processing Toolbox**: they binarized each image into “self” vs. “background” pixels and computed *self-size* — the proportion of the frame occupied by the person — as an indirect indicator of self-focus. They also **manually coded** smile intensity in each photo. Results showed that *awe* group participants progressively gave more space to the surroundings and smiled more intensely week over week.

1.2 Business Case

The original study’s measurement protocol has two practical limitations:

1. **Scalability**: manual smile coding is costly and subjective; MATLAB requires session-by-session manual thresholding.
2. **Emotional richness**: *self-size* captures framing behavior, not emotion directly; manual coding only distinguishes smile intensity without mapping the full emotional spectrum.

A natural extension of the study would be to **replace both metrics with an automatic facial expression classifier**. Such a classifier would enable:

- Automatic tracking of **longitudinal emotional trajectories** (week 1 → week 8).
- Comparison of emotion distributions between the *awe* and control groups over time.
- Testing the hypothesis that the *awe* group will show a progressive increase in **happy** classifications and a decrease in **neutral/sad**, consistent with the original study’s findings.
- Scaling the protocol to studies with more participants without a proportional increase in analysis cost.

The selfies collected in the original study are not publicly available — the participants are older adults and the data is kept under ethical and privacy restrictions. Therefore, this work trains the classifier on the public **FER-2013** dataset as a *proxy*. FER-2013 offers the same classification task (7 emotions, facial images under variable conditions) and allows

validating the architecture before its eventual application to the study’s corpus.

1.3 Technical Objective

Train a **Convolutional Neural Network (CNN)** capable of classifying human facial images into **7 emotion categories**: *angry*, *disgust*, *fear*, *happy*, *neutral*, *sad*, and *surprise*.

2 Steps and Selected Parameters

2.1 Dataset: FER-2013

Table 1: FER-2013 dataset overview

Attribute	Value
Source	Kaggle — <code>msambare/fer2013</code>
Total images	$\approx 35,000$
Training set	28,709 images
Test set	7,178 images
Resolution	48×48 px, grayscale
Classes	7 (angry, disgust, fear, happy, neutral, sad, surprise)

2.2 Preprocessing and Data Augmentation

Images were normalized by dividing pixel values by 255 to scale them to the $[0, 1]$ range. Data augmentation was applied to the training set to improve generalization and reduce overfitting:

```

1 train_datagen = ImageDataGenerator(
2     rescale=1./255,
3     rotation_range=10,
4     width_shift_range=0.1,
5     height_shift_range=0.1,
6     horizontal_flip=True,
7     zoom_range=0.1
8 )
9 test_datagen = ImageDataGenerator(rescale=1./255)

```

The test set receives only normalization, with no random transformations, to ensure a reproducible evaluation.

2.3 CNN Architecture

The network consists of three convolutional blocks followed by classification layers:

Table 2: CNN architecture

Layer	Configuration	Parameters
Conv2D (block 1)	32 filters, 3×3 kernel, ReLU	320
BatchNormalization	—	128
MaxPooling2D	2×2 pool	0
Conv2D (block 2)	64 filters, 3×3 kernel, ReLU	18,496
BatchNormalization	—	256
MaxPooling2D	2×2 pool	0
Conv2D (block 3)	128 filters, 3×3 kernel, ReLU	73,856
BatchNormalization	—	512
MaxPooling2D	2×2 pool	0
Flatten	—	0
Dense	256 units, ReLU	524,544
Dropout	rate = 0.5	0
Dense (output)	7 units, Softmax	1,799
Total trainable parameters		619,463

2.4 Compilation and Training

```

1 model.compile(
2     optimizer='adam',
3     loss='categorical_crossentropy',
4     metrics=['accuracy']
5 )
6 model.fit(train_generator, epochs=50,
7         validation_data=test_generator, callbacks=callbacks)

```

- **Optimizer:** Adam — adapts the learning rate per parameter, accelerating convergence in image classification tasks.
- **Loss function:** `categorical_crossentropy` — appropriate for multi-class classification with one-hot encoded labels.
- **Maximum epochs:** 50 (controlled by callbacks).
- **Batch size:** 64.

2.4.1 Training Callbacks

Three callbacks were used to improve training quality:

Table 3: Configured training callbacks

Callback	Key parameters	Function
ModelCheckpoint	monitor='val_accuracy', save_best_only=True	Saves <code>best_model.keras</code> whenever <code>val_accuracy</code> improves.
ReduceLROnPlateau	factor=0.5, patience=3, min_lr=1e-6	Halves the learning rate if <code>val_loss</code> does not improve for 3 consecutive epochs.
EarlyStopping	patience=8, restore_best_weights=True	Stops training if <code>val_loss</code> does not improve for 8 epochs; restores the best weights automatically.

3 Results

3.1 Global Metrics

The model achieved a **global accuracy of 63.03%** on the test set (loss: 0.9993).

3.2 Per-Class Classification Report

Table 4: Classification metrics by emotion

Emotion	Precision	Recall	F1-Score	Support
Angry	0.52	0.58	0.55	958
Disgust	0.80	0.18	0.29	111
Fear	0.49	0.32	0.38	1024
Happy	0.85	0.86	0.85	1774
Neutral	0.56	0.69	0.62	1233
Sad	0.49	0.50	0.50	1247
Surprise	0.76	0.77	0.76	831
Macro avg	0.64	0.55	0.56	7178
Weighted avg	0.63	0.63	0.62	7178

3.3 Training Curves

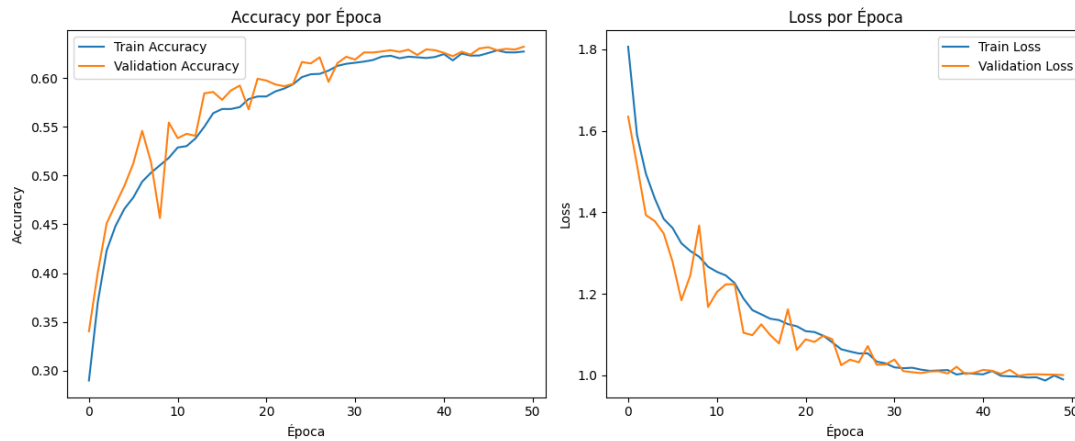


Figure 1: Accuracy and loss per epoch during training

3.4 Confusion Matrix

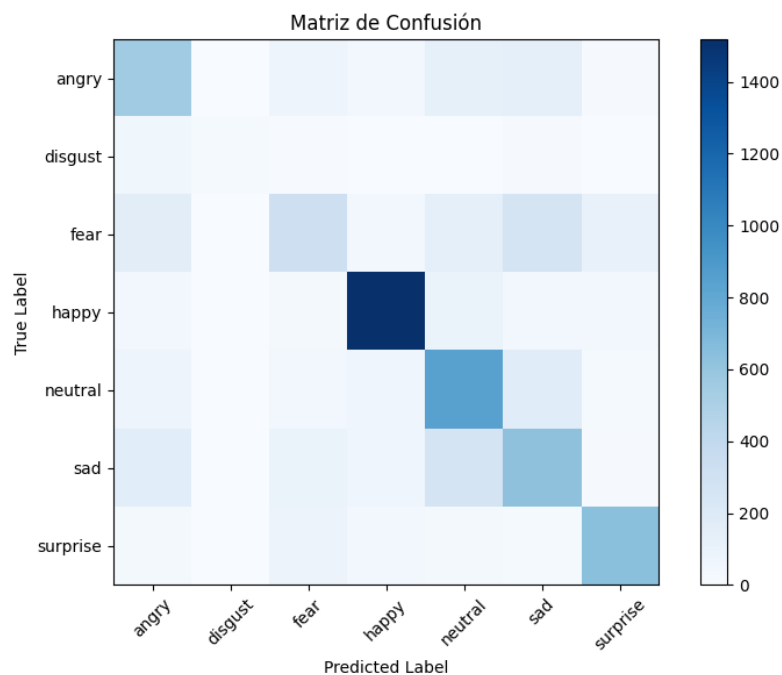


Figure 2: Confusion matrix on the test set

4 Conclusions

4.1 Per-class performance

The model achieved 63% global accuracy on FER-2013, with highly variable per-class performance. *Happy* obtained the best result ($F1 = 0.85$), thanks to its highly distinctive facial features. *Surprise* also performed well ($F1 = 0.76$). In contrast, *disgust* achieved

the lowest F1 (0.29) due to severe class imbalance: only 111 test samples versus 1,774 for *happy*.

4.2 Most difficult emotions

Fear and *sad* showed the weakest performance ($F1 = 0.38$ and 0.50 , respectively) for two complementary reasons: they share similar microexpressions — downturned mouth corners, tense eyebrows — and class imbalance prevents the model from learning sufficiently discriminative representations for *disgust*.

4.3 Relevance to the business case

From the perspective of Sturm et al., the most important result is that *happy* — the central emotion of the *awe walks* protocol — achieves $F1 = 0.85$, validating the classifier for tracking longitudinal emotional trajectories. Strong performance on *surprise* ($F1 = 0.76$) is also relevant, since awe is frequently expressed as a combination of happiness and surprise. The weaker performance on *disgust* and *fear* is not a critical obstacle, as those emotions are not the primary hypotheses of interest in the study.

4.4 Proposed improvements

1. **Class balancing:** apply oversampling (SMOTE or aggressive augmentation) to minority classes, particularly *disgust*.
2. **Transfer learning:** replace the from-scratch architecture with a pre-trained backbone (VGG16, EfficientNetB0) and fine-tune. Such architectures reach 70–75% accuracy on FER-2013.
3. **Domain adaptation:** the study’s corpus consists of selfies of older adults outdoors, a different domain from FER-2013 (laboratory/internet images). A domain adaptation cycle or fine-tuning on a small annotated sample would significantly improve real-world performance.
4. **Per-class confidence thresholds:** in production, use differentiated thresholds to reject low-confidence predictions for classes such as *fear* and *disgust*.

References

- [1] Sturm, V. E., Datta, S., Roy, A. R. K., Sible, I. J., Kosik, E. L., Veziris, C. R., Chow, T. E., Morris, N. A., Neuhaus, J., Kramer, J. H., Miller, B. L., Holley, S. R., & Keltner, D. (2022). *Big smile, small self: Awe walks promote prosocial positive emotions in older adults*. *Emotion*, 22(5), 1044–1058. <https://doi.org/10.1037/emo0000876>

- [2] Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., . . . Bengio, Y. (2013). *Challenges in Representation Learning: A Report on Three Machine Learning Contests*. arXiv:1307.0414.
- [3] Sambare, M. (2020). *FER-2013* [Dataset]. Kaggle. <https://www.kaggle.com/datasets/msambare/fer2013>
- [4] LeCun, Y., Boser, B., Denker, J. S., Henderson, D., Howard, R. E., Hubbard, W., & Jackel, L. D. (1989). *Backpropagation applied to handwritten zip code recognition*. *Neural Computation*, 1(4), 541–551.
- [5] Abadi, M., et al. (2015). *TensorFlow: Large-scale machine learning on heterogeneous systems*. <https://www.tensorflow.org>